

Data and Analysis Limitations

GradCafe Applicant Analysis | Module 3

30,040	46.53%	38.56%	34.59%
DATABASE ROWS	INTERNATIONAL ENTRIES	AMERICAN ACCEPTANCE	INTERNATIONAL ACCEPTANCE

1. Self-Reporting and Selection Bias

The GradCafe dataset is a large collection of anonymous, self-reported entries rather than a random or complete sample of graduate-school applicants. People choose whether to report their results, and reporting behavior may differ by university, program, nationality, decision outcome, and application year. For example, international entries represent 46.53% of records with a reported applicant group, while the calculated Fall 2026 acceptance percentages are 38.56% for American entries and 34.59% for international entries. These percentages describe only the submitted GradCafe records and cannot establish the official acceptance rates of either population. Duplicate prevention by URL improves internal consistency, but it cannot correct selection bias, misreporting, or applicants who never submitted a result.

2. Missing Values and Field Reliability

Missing, inconsistent, and transformed values also limit the precision of the analysis. PostgreSQL currently contains 11,988 rows without a usable GPA, 29,086 without a valid GRE Quantitative score, 28,038 without a valid GRE Verbal score, and 28,233 without a valid Analytical Writing score. Out-of-range values were stored as SQL NULL rather than included in averages, which avoids obviously invalid calculations but reduces the available sample and may introduce additional bias if missingness is not random. The LLM-generated program and university fields improve search consistency but may contain normalization errors, and new records collected by the scraper remain NULL in those fields until a separate enrichment process is performed. The original and LLM fields both produced 28 matching records for the selected Computer Science PhD comparison, but agreement on that query does not prove that every generated value is accurate. Results should therefore be interpreted as descriptive patterns in the collected dataset, not authoritative admissions statistics.